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Antares Rockets:

Antares is an expendable launch system developed by Orbital Sciences Corporation (OSC).The main purpose of this rocket is to launch the Cygnus spacecraft to the International Space Station as a part of NASA's COTS and CRS programs. Antares is the largest rocket operated by orbital ATK.NASA awarded Orbital a Commercial Orbital Transportation Services. This rocket also provides cargo to the International Space Station. The height of this rocket is around 40.5m with a diameter of 3.9m.The minimum mass of this rocket is 2, 82,000kg. It has 2 to 3 stages for its launch. Its payload to LEO is 8,000kg.

S.no.	Year	No. of Rockets	Launch Outcome
.1	2013	2	Success
.2	2014	3,1	Success, Failure
.3	2015	N.A	N.A
.4	2016	1	Success
.5	2017	1	Success
.6	2018	2	Success
.7	2019	1	Success

Atlas V Rockets:

Atlas V is the fifth major version in the Atlas Rocket family. This expendable launch system was originally engineered by Lockheed Martin and is now operated by United Launch Alliance (ULA).This rocket consists of 2 main stages. The first stage is powered by Russian RD-180 engine and runs with the help of RD Amross and burning kerosene and liquid oxygen. The second stage is made up of the RL 10 engine made by US and uses liquid oxygen and liquid hydrogen to propel. This rocket is said to have AJ-60A strap-on boosters. This rocket helped in the transportation of cargo to space stations. This rocket has a height of 58.3m with a diameter of 3.81m and has a mass of 590,000kg.The payload to LEO is around 8,250kg and has a payload of around 4,750kg to GTO.

S.no	Date	Payload	Launch Outcome
1.	21st Aug 2002	Hot Bird 6	Success
2.	20th Apr 2006	New horizons	Success
3.	10th Dec 2007	USA-198	Success
4.	7th May 2011	CCAFS SLC-41	Success
5.	5th Aug 2011	CCAFS SLC-41	Success

Delta IV Rockets:

Delta IV is an expendable launch system in the Delta rocket family. The main components were designed by Boeing's Defense and Security division. Delta IV vehicles are currently built in the United Launch Alliance (ULA).This rocket can launch commercial satellite payloads and also heavy payloads. One of the main purpose of this rocket is to launch U.S. military Payloads. This rocket uses liquid hydrogen to propel for launch. The minimum height of this rocket is 63m to a maximum of 72m.It has a diameter of 5m and has a mass of approx. 250,000kg.The payload to LEO for this rocket is 11,470kg and has a payload of 4,440kg to GTO.

S. No	Date	Payload	Launch Outcome
1.	20-11-2002	Eutletusat W5	Success
2.	11-03-2003	Military Communications Satellite	Success
3.	21-12-2004	Demonstration Payload	Success
4.	16-07-2011	Navigation Satellite	Success
5.	17-05-2014	Navigation Satellite	Success

Falcon 9 Rockets:

Falcon 9 is a two staged medium lift launch vehicle designed and manufactured by SpaceX in the United States. It is made by Merlin engines. It had burning liquid nitrogen and rocket-grand kerosene

propellants. There are nine engines in the first stage of the rocket. It has also developed in its expendable launch systems, Falcon 9 is partially reusable, with the first stage capable of re-entering the atmosphere and landing back vertically. The Falcon 9 has a height of around 70m, has a diameter of 3.7m,has a mass of around 55,000kg and has 2 stages. Let's see information about the payload of this rocket. This rocket has a Payload of 22,800kg to LEO orbit, 8,300kg to GTO, 4,020kg to Mars (heliocentric orbit).

S.no	Year	No. of rockets	Launch Outcome
1.	2010	2	Success
2.	2011	0	Success
3.	2012	1,1	Success, Partial Failure
4.	2013	3	Success
5.	2014	6	Success
6.	2015	6,1	Success, Failure
7.	2016	8,1	Success
8.	2017	19	Success
9.	2018	21	Success
10.	2019	10	Success

Minotaur V Rockets:

Minotaur V is an American expendable launch system derived from the Minotaur IV. It was developed by Orbital Science Corporation (OSC) and made its maiden flight on 7th September. Minotaur V is a five staged vehicle and is designed to place up to 630kg of payload into GTO orbit. One of the main launch pads of this rocket is Space Launch Complex 8.This rocket has a height of 24.56m and a diameter of 2.34m with a mass of 89,373kg.Its payload to GTO is 532kg and a payload of 342kg to TLI. This rocket has a STAR-37 as a fifth stage. Two variants are available in the rocket.

Pegasus Rockets:

he Pegasus is an air launched rocket developed by Orbital Science Corporation (OSC).This rocket is capable of carrying small payloads of upto 443kg into low earth orbit (LEO).This rocket first flew in 1990 and still remains as an important rocket and is still used today. The rocket uses 3 solid propellant stages and an optional monopropellant fourth stage. The first stage has a wing and a tail to provide lift and attitude control of the rocket. The main reason why this rocket is named Pegasus as it has wings in the first stage and resembles of a horse with wings (Pegasus).This rocket can also by released from an aircraft carrier at approximately 40,000ft.The Pegasus's three Orion solid motors were developed by Hercules Aerospace specifically for the Pegasus Launcher. This was first launched in the United States. This rocket has a height of 16.9m with a diameter of 1.27m.The mass of this rocket is 18,500kg and this rocket has 3 stages

S.no	Year	No. of rockets	Launch Outcome
1.	1990	1	Success
2.	1995	2,1	Success, Failure
3.	1998	6	Success
4.	2003	4	Success
5.	2007	1	Success
6.	2016	1	Success